# Alpha-Driven RFQ Simulator Technical Specification

Version 2 — Jupyter Notebook Implementation Guide

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## Overview

This document specifies a discrete-event simulator for the alpha-driven liquidity provision strategy. The simulator models a single bond over a configurable horizon, generates RFQ flow with explicit competitor pricing, and runs both the liquidity-provision strategy and a naive-aggressor baseline in parallel for direct comparison.

The pipeline:

1. Generate a true price path with intraday seasonality (Section 3).
2. Generate an alpha signal with regime-dependent noise (Section 4).
3. Generate RFQ arrivals with intraday seasonality (Section 6).
4. Simulate explicit competitor quotes per RFQ (Section 9).
5. At each RFQ, compute the optimal quote, resolve the fill against competitors (Sections 10–11).
6. Execute hybrid exit logic when alpha decays (Section 12).
7. Optionally apply hedging overlays (Section 14).
8. Track inventory, P&L, and diagnostics (Section 15).
9. Run the naive-aggressor baseline on the same paths (Section 16).

All random processes are seeded. The simulator is single-name for V1; multi-name portfolio effects are deferred.

## Simulation Clock

| | | |
|---|---|---|
| `T_days` | int | Horizon in trading days (default: 60) |
| `dt_minutes` | float | Price step size (default: 5) |
| `trading_hours` | float | Hours per trading day (default: 8) |
| `seed` | int | Master random seed (default: 42) |

Clock runs in continuous minutes from $t = 0$ to $t = T_{\text{days}} \times 480$. Price discretized at `dt_minutes`. RFQ arrivals at arbitrary times within trading hours.

# True Price Process

Arithmetic Brownian motion with mean-reverting drift and intraday volatility seasonality:

$$p_{t+dt} = p_t + \kappa(\bar{p} - p_t)\,dt + \sigma(h)\sqrt{dt}\,\epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0,1) \tag{1}$$

where $h$ is the hour within the trading day and the intraday vol profile is:

$$\sigma(h) = \sigma_{\text{base}} \cdot \left(1 + v_{\text{open}} \cdot e^{-(h-h_0)^2/\tau_v^2} + v_{\text{close}} \cdot e^{-(h-h_c)^2/\tau_v^2}\right) \tag{2}$$

This produces higher volatility at open and close (U-shape), matching empirical bond price dynamics.

**Unit conversion:**

$$\sigma_{\text{per step}} = \frac{\sigma_{\text{daily,bps}}}{10{,}000} \times p_0 \times \sqrt{\frac{dt_{\text{min}}}{480}} \tag{3}$$

| | | |
|---|---|---|
| `p0` | float | Initial price (default: 100.0) |
| `sigma_daily_bps` | float | Daily vol in bps (default: 50) |
| `kappa_daily` | float | Mean-reversion speed (default: 0.02) |
| `p_bar` | float | Long-run mean price (default: 100.0) |
| `v_open` | float | Open vol multiplier (default: 0.5) |
| `v_close` | float | Close vol multiplier (default: 0.3) |
| `tau_v_hours` | float | Vol seasonality width (default: 0.75) |

IG corporates: $\sigma_{\text{daily}} \in [20, 80]$ bps, $\kappa \in [0.005, 0.05]$. HY corporates: $\sigma_{\text{daily}} \in [50, 200]$ bps, $\kappa \in [0.01, 0.10]$.

# Alpha Signal

## Signal Generation

The theoretical alpha at time $t$ is the true forward return over horizon $H$:

$$\alpha_t^* = p_{t+H} - p_t \tag{4}$$

The trader observes a noisy version:

$$\alpha_t = \rho(r_t) \cdot \alpha_t^* + \sqrt{1 - \rho(r_t)^2}\, \sigma_\alpha\, \eta_t, \qquad \eta_t \sim \mathcal{N}(0,1) \tag{5}$$

where $\rho(r_t)$ is the regime-dependent IC (see below) and $\sigma_\alpha = \mathrm{Std}(\alpha^*)$ computed from the path.

## Regime-Dependent IC

The signal quality varies with a latent volatility regime. Define a two-state Markov chain $r_t \in \{\text{calm}, \text{stressed}\}$ with daily transition matrix:

$$\mathbf{P} = \begin{pmatrix} 1 - p_{cs} & p_{cs} \\ p_{sc} & 1 - p_{sc} \end{pmatrix} \tag{6}$$

In the calm regime, IC is at baseline. In the stressed regime, IC degrades (alpha models perform worse in vol spikes):

$$\rho(r_t) = \begin{cases} \mathrm{IC} & r_t = \text{calm} \\ \mathrm{IC} \cdot \mathrm{IC_stress_mult} & r_t = \text{stressed} \end{cases} \tag{7}$$

## Signal Decay and Refresh

Linear decay: $\alpha_{\text{rem}}(t) = \alpha_{t_0} \cdot \max(0,\ (t_0 + H - t)/H)$.

New signals generated every `signal_refresh_hours`.

| | | |
|---|---|---|
| `IC` | float | Baseline information coefficient (default: 0.10) |
| `IC_stress_mult` | float | IC multiplier in stressed regime (default: 0.4) |
| `alpha_horizon_days` | float | Signal horizon $H$ (default: 10) |
| `signal_refresh_hours` | float | Update frequency (default: 24) |
| `p_calm_to_stress` | float | Daily transition prob (default: 0.05) |
| `p_stress_to_calm` | float | Daily transition prob (default: 0.15) |

Credit alpha: IC $\in [0.03, 0.15]$, horizons 5–20 days. Stressed IC is typically 30–60% of calm IC. With $p_{cs} = 0.05$, $p_{sc} = 0.15$, the stationary probability of stress is 25% and average stress duration is $\sim$7 days.

# Target Position

$$q_t^* = \text{clip}\left(\frac{\alpha_{\text{rem}}(t)}{\gamma \cdot \sigma^2_{\text{daily}}},\ -q_{\max},\ q_{\max}\right) \tag{8}$$

Inventory in lots (each lot = `lot_size_mm` \$MM face). The clip enforces hard limits. A soft penalty zone is applied near limits via the lean convexity term (Section 8).

Feasibility constraint (diagnostic, not binding):

$$q^*_{\text{feasible}} \leq \mu_{\text{fav}} \cdot \bar{P}(\text{win}) \cdot (T - t) \tag{9}$$

If $q^*$ exceeds feasible, log a warning. The trader may still target it but should expect under-delivery.

| | | |
|---|---|---|
| `gamma` | float | Risk aversion (default: 1.0) |
| `q_max` | int | Max position in lots (default: 20) |
| `lot_size_mm` | float | \$MM per lot (default: 1.0) |

# RFQ Arrival Process

## Arrival Times with Intraday Seasonality

RFQs arrive as an inhomogeneous Poisson process. The instantaneous rate at hour $h$ within the trading day:

$$\mu(h) = \mu_{\text{base}} \cdot f(h), \qquad f(h) = 1 + A_{\text{open}} \cdot e^{-(h-h_0)^2/\tau_f^2} + A_{\text{close}} \cdot e^{-(h-h_c)^2/\tau_f^2} \tag{10}$$

where $h_0$ and $h_c$ are the opening and closing hours, and $\mu_{\text{base}}$ is calibrated so the daily integral $\int f(h)\,dh$ gives the desired `rfq_rate_per_day`.

**Simulation via thinning.** Generate candidate arrivals from a homogeneous Poisson with rate $\mu_{\max} = \mu_{\text{base}} \cdot \max_h f(h)$. Accept each candidate at time $h$ with probability $f(h)/\max_h f(h)$.

## RFQ Attributes

Each RFQ is characterized by:

**Direction:** $P(\text{client buy}) = 0.5 + \delta_{\text{flow}}$.

**Size:** size $= \lceil \exp(\mu_s + \sigma_s Z) \rceil$, clipped to $[1, \text{size_max}]$.

**Dealer count:** $N = 1 + \text{Poisson}(\bar{N} - 1)$, clipped to $[1, N_{\max}]$.

**Client toxicity:** $\tau_c \sim \text{Beta}(a_{\text{tox}}, b_{\text{tox}})$.

| | | |
|---|---|---|
| `rfq_rate_per_day` | float | Mean daily RFQs (default: 15) |
| `A_open` | float | Open activity multiplier (default: 0.8) |
| `A_close` | float | Close activity multiplier (default: 0.5) |
| `tau_f_hours` | float | Seasonality width (default: 0.75) |
| `flow_bias` | float | $\delta_{\text{flow}} \in [-0.3, 0.3]$ (default: 0.0) |
| `size_mu` | float | Log-mean of size (default: 0.5) |
| `size_sigma` | float | Log-std of size (default: 0.8) |
| `size_max` | int | Max size in lots (default: 10) |
| `n_dealers_mean` | float | Mean competitors $\bar{N}$ (default: 4) |
| `n_dealers_max` | int | Max competitors (default: 8) |
| `tox_a` | float | Beta shape $a$ (default: 2.0) |
| `tox_b` | float | Beta shape $b$ (default: 8.0) |

Liquid IG: 15–40 RFQs/day, 4–5 dealers, sizes 1–5MM. Illiquid HY: 2–8 RFQs/day, 2–3 dealers, sizes 0.5–3MM. Beta(2, 8) $\rightarrow$ mean toxicity 0.2, 90th pctile $\approx$ 0.45.

## Observable Mid and Skew

The trader observes a lagged, noisy mid:

$$\text{mid}_{\text{obs},t} = p_{t-\Delta t_{\text{lag}}} + \sigma_{\text{obs}} \cdot \xi_t, \qquad \xi_t \sim \mathcal{N}(0,1) \tag{11}$$

The skew is a noisy correction for staleness:

$$s_t = \rho_s \cdot (p_t - \text{mid}_{\text{obs},t}) + \sqrt{1-\rho_s^2}\, \sigma_{\text{stale}}\, \zeta_t \tag{12}$$

Trader's theo: $\text{theo}_t = \text{mid}_{\text{obs},t} + s_t + \ell(q_t, q_t^*)$.

| | | |
|---|---|---|
| `obs_lag_minutes` | float | Pricing staleness (default: 15) |
| `obs_noise_bps` | float | Observation noise (default: 5) |
| `skew_accuracy` | float | $\rho_s$ (default: 0.3) |

## Lean Computation

$$\ell(q_t, q_t^*) = -\lambda_{\text{eff}} \cdot (q_t - q_t^*) \tag{13}$$

$$\lambda_{\text{eff}} = \lambda_{\text{base}} \cdot \underbrace{\left(\frac{\sigma_{\text{daily}}}{\sigma_{\text{ref}}}\right)^2}_{\text{vol}} \cdot \underbrace{\left(1 + \kappa_u \cdot \frac{t - t_{\text{sig}}}{H}\right)}_{\text{urgency}} \cdot \underbrace{\phi(q_t, q_t^*)}_{\text{convexity + soft limit}} \tag{14}$$

### Soft Limit Penalty Zone

The convexity function incorporates a soft penalty as inventory approaches $q_{\max}$:

$$\phi(q_t, q_t^*) = \big(1 + \kappa_c \cdot |q_t - q_t^*|\big) \cdot \big(1 + \kappa_{\text{lim}} \cdot \max(0,\ |q_t|/q_{\max} - \theta_{\text{lim}})^2\big) \tag{15}$$

The second factor activates when inventory exceeds fraction $\theta_{\text{lim}}$ of the hard limit, applying a quadratic penalty that makes the lean increasingly aggressive near the boundary. For example, with $\theta_{\text{lim}} = 0.7$ and $\kappa_{\text{lim}} = 10$, lean doubles at 80% capacity and quadruples at 90%.

| | | |
|---|---|---|
| `lambda_base_bps` | float | Base lean per lot gap (default: 2.0) |
| `sigma_ref_bps` | float | Reference vol for normalization (default: 50) |
| `kappa_urgency` | float | Urgency scale at signal expiry (default: 1.0) |
| `kappa_convexity` | float | Convexity per lot of gap (default: 0.1) |
| `kappa_limit` | float | Soft limit penalty strength (default: 10.0) |
| `theta_limit` | float | Fraction of $q_{\max}$ where penalty activates (default: 0.7) |

## Win-Rate: Explicit Competitor Model

Rather than a parametric logistic approximation, V2 simulates competitors explicitly. This is the highest-priority modeling component.

## Competitor Pricing Model

Each competing dealer $j \in \{1, \ldots, N-1\}$ generates a quote around the true price with idiosyncratic bias and noise:

$$p_j = p_{\text{true}} + \underbrace{b_j}_{\text{dealer bias}} + \underbrace{m_j(\mathbf{z})}_{\text{markup}} + \underbrace{\epsilon_j}_{\text{noise}} \tag{16}$$

**Dealer bias** $b_j$: each dealer has a persistent lean reflecting their own inventory. Drawn once per simulation day:

$$b_j \sim \mathcal{N}(\bar{b}_{\text{street}},\ \sigma_b) \tag{17}$$

where $\bar{b}_{\text{street}}$ is the aggregate street lean (see Section 13) and $\sigma_b$ is cross-dealer dispersion.

**Markup** $m_j(\mathbf{z})$: the dealer's intended edge, varying by context:

$$m_j(\mathbf{z}) = \bar{m} + m_N \cdot (N - \bar{N}) + m_{\text{size}} \cdot \log(\text{size}) + m_{\text{tox}} \cdot \tau_c + u_j \tag{18}$$

where $u_j \sim \mathcal{N}(0, \sigma_m)$ is per-quote noise and the coefficients capture competitive dynamics: more dealers $\rightarrow$ tighter markups ($m_N < 0$), larger size $\rightarrow$ wider ($m_{\text{size}} > 0$), toxic client $\rightarrow$ wider ($m_{\text{tox}} > 0$).

**Quote noise** $\epsilon_j \sim \mathcal{N}(0, \sigma_q)$: captures model error and information differences across dealers.

**Direction convention.** For a client-buy RFQ, all dealer quotes are offers (higher = worse for dealer). The trader wins if their offer is the lowest:

$$\text{filled} = \mathbb{1}\left[p_{\text{trader}} < \min_j p_j\right] \tag{19}$$

For client-sell, quotes are bids (higher = better for client); trader wins if bid is highest.

## Dealer Participation

Each dealer participates with probability:

$$P_j(\text{respond}) = \bar{r} - r_{\text{size}} \cdot \log(\text{size}) - r_{\text{tox}} \cdot \tau_c \tag{20}$$

clipped to $[0.3, 1.0]$. Non-responding dealers are removed from the competition for that RFQ. This captures the real-world pattern where fewer dealers respond to large or toxic tickets.

| | | |
|---|---|---|
| `dealer_bias_std_bps` | float | $\sigma_b$, cross-dealer lean dispersion (default: 3.0) |
| `markup_base_bps` | float | $\bar{m}$, baseline dealer markup (default: 8.0) |
| `markup_N_bps` | float | $m_N$, markup shift per dealer (default: $-1.0$) |
| `markup_size_bps` | float | $m_{\text{size}}$, shift per log-lot (default: 2.0) |
| `markup_tox_bps` | float | $m_{\text{tox}}$, shift per unit toxicity (default: 10.0) |
| `markup_noise_bps` | float | $\sigma_m$, per-quote markup noise (default: 3.0) |
| `quote_noise_bps` | float | $\sigma_q$, dealer model error (default: 2.0) |
| `respond_base` | float | $\bar{r}$, base response rate (default: 0.85) |
| `respond_size` | float | $r_{\text{size}}$, size impact on response (default: 0.05) |
| `respond_tox` | float | $r_{\text{tox}}$, toxicity impact on response (default: 0.1) |

The effective win-rate curve implied by the competitor model should be validated: simulate 10,000 RFQs at varying trader markups (holding all else fixed), plot the empirical win rate vs. markup, and verify it produces a smooth sigmoid with center and steepness matching the target regime (IG: center 3–8bps, steep; HY: center 15–40bps, flat). Adjust $\bar{m}$ and $\sigma_q$ to shift and flatten the curve.

## Optimal Quoting

Since the trader does not know competitor quotes in advance, the trader still optimizes against an *estimated* win-rate model. However, the fill decision is resolved against the explicitly simulated competitors.

### Trader's Internal Win-Rate Estimate

The trader uses a logistic model (calibrated offline against historical simulated competitors):

$$\hat{P}(\text{win} \mid m, \mathbf{z}) = \frac{1}{1 + \exp(-\hat{\beta}(\mathbf{z}) \cdot (\hat{c}(\mathbf{z}) - |m|))} \tag{21}$$

where $m$ is the markup over theo and $\hat{c}$, $\hat{\beta}$ are the trader's estimates of the competitive landscape. These can be initialized from the true competitor parameters, then optionally degraded to simulate estimation error.

### Optimization

The trader maximizes:

$$m^* = \arg\max_m \; \hat{P}(\text{win} \mid m, \mathbf{z}) \cdot \big[\text{edge}(m) + \Delta V\big] \tag{22}$$

**Continuation value approximation:**

$$\Delta V = \begin{cases} +\alpha_{\text{rem}} \cdot |\Delta q| & \text{if favorable (toward } q^*) \\ -\gamma \cdot \sigma^2 \cdot |q_t - q_t^*| \cdot |\Delta q| / q_{\max} & \text{if adverse (away from } q^*) \end{cases} \tag{23}$$

Solve on a grid $m \in [-m_{\max}, +m_{\max}]$ at 0.1bps resolution.

### Quote Bounds

1. **Alpha bound**: $\text{edge}(m) + \Delta V \geq -|\alpha_{\text{rem}}| \cdot |\Delta q|$.
2. **Position bound**: $|q + \Delta q| \leq q_{\max}$.

If no feasible $m$ yields positive expected value, the trader **declines**.

### Fill Resolution

The trader's actual quote price $p_{\text{trader}} = \text{theo} + m^* \cdot \text{direction}$ is compared against the simulated competitor quotes:

$$\text{filled} = \begin{cases} \mathbb{1}[p_{\text{trader}} < \min_j p_j] & \text{client buy (offers)} \\ \mathbb{1}[p_{\text{trader}} > \max_j p_j] & \text{client sell (bids)} \end{cases} \tag{24}$$

This is the key improvement over V1: the fill outcome depends on the realized competitive landscape, not a parametric approximation.

| | | |
|---|---|---|
| `winrate_estimation_error` | float | Degradation factor on trader's $\hat{c}$, $\hat{\beta}$ (default: 0.0) |
| `m_max_bps` | float | Markup grid extent (default: 30) |
| `m_grid_bps` | float | Grid resolution (default: 0.1) |

## Fill Dynamics

### Post-Fill Adverse Move

After a fill, the true price jumps in the direction of the informed client's trade:

$$p_{t^+} = p_t + \text{direction} \cdot \tau_c \cdot \sigma_{\text{adverse}} \cdot |Z|, \qquad Z \sim \mathcal{N}(0,1) \tag{25}$$

This one-time jump is applied to the price path. Non-toxic clients ($\tau_c \approx 0$) produce negligible jumps.

| | | |
|---|---|---|
| `adverse_move_bps` | float | Adverse selection scale (default: 5.0) |

## Hybrid Exit Logic

When the alpha signal decays, the strategy transitions from patient liquidity provision to urgent aggressor execution. The exit operates in two phases controlled by a time-based trigger.

### Phase 1: Patient Exit via RFQ (Lean Reversal)

As $\alpha_{\text{rem}} \to 0$, the target $q^* \to 0$ and the lean naturally reverses to encourage position reduction via incoming RFQ flow. During this phase the trader earns spread on unwinding fills, exactly like the entry phase but in reverse.

### Phase 2: Aggressive Exit (Cross Spread)

If the position has not been sufficiently unwound by the deadline, the trader switches to aggressor mode. Define:

$$\text{exit_mode} = \begin{cases} \text{RFQ (patient)} & \text{if } t < t_{\text{sig}} + H - \Delta t_{\text{aggress}} \\ \text{aggressor} & \text{if } t \geq t_{\text{sig}} + H - \Delta t_{\text{aggress}} \end{cases} \tag{26}$$

In aggressor mode, the remaining position $q_t$ is unwound over the remaining time at a constant rate, paying the spread on each execution:

$$\text{aggress_cost_per_lot} = c_{\text{aggress}} + c_{\text{impact}} \cdot \sqrt{|q_t|} \tag{27}$$

The first term is the half-spread cost. The second captures market impact for larger positions (square-root impact model).

**Simulation.** During the aggressor window, at each time step, the simulator executes $\lceil |q_t| / n_{\text{remaining_steps}} \rceil$ lots at cost $p_t \pm \text{aggress_cost}$ and updates inventory.

| | | |
|---|---|---|
| `aggress_window_hours` | float | $\Delta t_{\text{aggress}}$, time before signal expiry to start aggressing (default: 8, i.e. 1 |
| `aggress_halfspread_bps` | float | $c_{\text{aggress}}$, half-spread cost (default: 8.0) |
| `aggress_impact_bps` | float | $c_{\text{impact}}$, impact per $\sqrt{\text{lot}}$ (default: 2.0) |

The aggressor window should be logged as a separate cost center in the P&L decomposition, so the user can see exactly how much the "impatient exit" costs relative to the "patient" RFQ approach. This is a key output for calibrating `aggress_window_hours`.

## Street Lean Inference

The aggregate dealer lean $\bar{b}_{\text{street}}$ is a latent variable that the trader can partially infer from observables. This section models both the true process and the trader's estimator.

### True Street Lean Process

The street lean follows a mean-reverting OU process:

$$d\bar{b} = \theta_b(\bar{b}_{\text{eq}} - \bar{b})\,dt + \sigma_b\,dW_t \tag{28}$$

Discretized daily. When $\bar{b} > 0$, the street is collectively long and leaning offers down (wanting to sell). This feeds into the competitor bias terms (Section 9).

### Observable Signatures

The trader estimates street lean from three noisy proxies:

**(i) Bid-ask asymmetry from received quotes.** When the trader responds to RFQs and loses, the trader sometimes observes the winning price (via TRACE with a delay). The ratio of offer-side vs. bid-side competitiveness reveals street lean:

$$\hat{b}_{\text{asym},t} = \text{EMA}\left(\frac{\bar{m}_{\text{offer}} - \bar{m}_{\text{bid}}}{\bar{m}_{\text{offer}} + \bar{m}_{\text{bid}}},\ \alpha_{\text{ema}}\right) \tag{29}$$

**(ii) Flow imbalance from TRACE.** Net signed volume over a rolling window, normalized:

$$\hat{b}_{\text{flow},t} = \frac{\sum_{\tau\in[t-w,t]} \text{signed_vol}_\tau}{\sum_\tau |\text{vol}_\tau|} \tag{30}$$

**(iii) ETF premium/discount.** Modeled as a noisy linear function of street lean:

$$\text{ETF_pd}_t = \phi_0 + \phi_1 \cdot \bar{b}_t + \nu_t, \qquad \nu_t \sim \mathcal{N}(0, \sigma_\nu) \tag{31}$$

**Combined estimator:** weighted average of the three proxies:

$$\hat{b}_t = w_1 \hat{b}_{\text{asym}} + w_2 \hat{b}_{\text{flow}} + w_3 \hat{b}_{\text{etf}} \tag{32}$$

The trader can use $\hat{b}_t$ as an additional input to the skew model: if the street is long (positive $\hat{b}$), the clearing price will eventually fall as dealers unwind, so skew negative.

| | | |
|---|---|---|
| `street_lean_mean_rev` | float | $\theta_b$ (default: 0.1) |
| `street_lean_vol_bps` | float | $\sigma_b$ (default: 2.0) |
| `street_lean_eq` | float | $\bar{b}_{\text{eq}}$ (default: 0.0) |
| `street_obs_noise` | float | Noise on each proxy (default: 0.5) |
| `street_proxy_weights` | 3-vec | $[w_1, w_2, w_3]$ (default: $[0.4, 0.4, 0.2]$) |

## Hedging Overlay

While waiting for favorable RFQ flow, the trader carries unhedged directional risk. An optional hedging overlay reduces this risk at the cost of basis risk and hedge execution costs.

### Hedge Instruments

Three hedging channels, each modeled as a correlated but imperfect offset:

**CDS hedge.** Offsets spread risk. Correlation with bond price: $\rho_{\text{cds}}$. Execution cost: `cds_cost_bps` per unit notional. Basis risk = $(1 - \rho_{\text{cds}}^2)$ fraction of variance remains.

**ETF hedge.** Offsets systematic credit. Correlation: $\rho_{\text{etf}}$. Cost: `etf_cost_bps`.

**Treasury hedge.** Offsets duration. Correlation: $\rho_{\text{tsy}}$. Cost: `tsy_cost_bps`.

### Hedge Ratio and Rebalance

The hedge position tracks the bond inventory:

$$h_t^{(k)} = -\delta_k \cdot q_t \cdot \beta_k \tag{33}$$

where $\delta_k \in \{0, 1\}$ is a toggle for each hedge instrument and $\beta_k$ is the hedge ratio (e.g., DV01-matched for rates, CS01-matched for CDS). The hedge is rebalanced at the end of each day.

Intraday, the hedge is static and the residual unhedged P&L accrues.

### Impact on Lean

When hedging is active, the effective inventory for lean purposes becomes the *residual* exposure:

$$q_{\text{eff}} = q_t \cdot \sqrt{1 - R^2_{\text{hedge}}} \tag{34}$$

where $R^2_{\text{hedge}}$ is the fraction of variance explained by the hedges. A well-hedged position requires less aggressive lean.

| | | |
|---|---|---|
| `hedge_cds_on` | bool | Enable CDS hedging (default: False) |
| `hedge_etf_on` | bool | Enable ETF hedging (default: False) |
| `hedge_tsy_on` | bool | Enable Treasury hedging (default: False) |
| `rho_cds` | float | CDS–bond correlation (default: 0.85) |
| `rho_etf` | float | ETF–bond correlation (default: 0.70) |
| `rho_tsy` | float | Treasury–bond correlation (default: 0.50) |
| `cds_cost_bps` | float | CDS execution cost (default: 2.0) |
| `etf_cost_bps` | float | ETF execution cost (default: 1.0) |
| `tsy_cost_bps` | float | Treasury execution cost (default: 0.5) |

## Accounting and P&L

### P&L Decomposition

$$\boxed{\text{PnL}_{\text{total}} = \text{PnL}_{\text{alpha}} + \text{PnL}_{\text{spread}} + \text{PnL}_{\text{carry}} + \text{PnL}_{\text{hedge}} - \text{Cost}_{\text{aggress}}} \tag{35}$$

Each component:

$$\text{PnL}_{\text{alpha}} = \textstyle\sum_t q_{t-1} \cdot (p_t - p_{t-1}) \cdot \text{lot_size} \tag{36}$$

$$\text{PnL}_{\text{spread}} = \textstyle\sum_{k \in \text{RFQ fills}} (p_k - p_{\text{true},k}) \cdot \text{signed_size}_k \cdot \text{lot_size} \tag{37}$$

$$\text{PnL}_{\text{carry}} = \textstyle\sum_{\text{days}} q_t \cdot \text{lot_size} \cdot \text{coupon_bps}/360 \tag{38}$$

$$\text{PnL}_{\text{hedge}} = \textstyle\sum_t h_t \cdot \Delta p_t^{(\text{hedge})} - \text{hedge_txn_costs} \tag{39}$$

$$\text{Cost}_{\text{aggress}} = \textstyle\sum_{k \in \text{aggress fills}} \text{spread_paid}_k + \text{impact_cost}_k \tag{40}$$

The aggressor cost is tracked separately so the user can directly measure the price of impatient exit.

| | | |
|---|---|---|
| `coupon_bps` | float | Annual coupon (bps of face, default: 500) |

## Naive Aggressor Baseline

Run a second strategy on the **identical price path and alpha signals**. The baseline trader uses the same $q_t^*$ but instead of providing liquidity via RFQ, immediately crosses the spread to reach target.

### Execution Model

At each signal refresh (or whenever $q^*$ changes by more than a threshold), the baseline trader executes $\Delta q = q_t^* - q_t$ at cost:

$$\text{cost_per_lot} = c_{\text{aggress}} + c_{\text{impact}} \cdot \sqrt{|\Delta q|} \tag{41}$$

using the same aggressor cost parameters as the hybrid exit (Section 12). The baseline trader has no RFQ interaction, no lean, no win-rate optimization.

### Comparison Metrics

The following should be reported side-by-side:

| Metric | LP Strategy | Aggressor Baseline |
|---|---|---|
| Total P&L | Full decomposition | Alpha + carry − execution costs |
| Alpha Capture Ratio | Typically 0.7–0.8 | ∼1.0 (immediate execution) |
| Net Execution Cost | Often negative (earning spread) | Always positive |
| Sharpe Ratio | Higher if spread income > ACR loss | Lower |
| Max Drawdown | Compare | Compare |

The key question the comparison answers: **does the spread income earned by the LP strategy exceed the alpha lost from slower position convergence?** If yes, the strategy dominates the baseline.

For slow alpha (horizon $\geq$ 10 days) and expensive execution (half-spread $\geq$ 5bps), the LP strategy should dominate. For fast alpha (horizon $\leq$ 3 days) or cheap execution (half-spread $\leq$ 2bps), the baseline may be competitive because the cost of delayed convergence exceeds the spread savings.

# Diagnostics and Output

## Time Series Plots

1. **Price panel**: true $p_t$, observable mid, theo. Mark fill events (triangles: favorable green, adverse red).
2. **Inventory panel**: $q_t$ and $q_t^*$, shaded by regime (calm/stressed). Mark aggressor exit window.
3. **Cumulative P&L**: stacked area — alpha, spread, carry, hedge, aggressor cost. Overlay baseline total P&L.
4. **Street lean**: true $\bar{b}_t$ and trader's estimate $\hat{b}_t$.

## RFQ Event Log (DataFrame)

| | |
|---|---|
| `time, direction, size, n_dealers, toxicity, favorable` | RFQ attributes |
| `p_true, mid_obs, skew, lean, theo` | Pricing stack |
| `optimal_markup, quote, win_prob_est` | Trader's decision |
| `best_competitor, filled, spread_pnl` | Outcome (from explicit competitor sim) |
| `q_before, q_after, declined, regime` | State |

## Summary Statistics

Total P&L, alpha P&L, spread P&L, hedge P&L, aggressor cost, ACR, Sharpe, fill rate (favorable/adverse), decline rate, convergence time, adverse selection (1h MTM post-fill), max drawdown. All reported for both LP strategy and baseline.

# Scenario Sweep

Priority sweep dimensions:

| Dimension | Range | Question |
|---|---|---|
| `lambda_base_bps` | [0.5, 1, 2, 4, 8] | Optimal lean aggressiveness? |
| `IC` | [0.03, 0.05, 0.10, 0.15] | Minimum viable alpha quality? |
| `rfq_rate_per_day` | [3, 8, 15, 30] | Does strategy work in illiquid names? |
| `markup_base_bps` | [3, 8, 15, 30] | Effect of dealer competitiveness? |
| `aggress_window_hours` | [0, 4, 8, 16] | How much patience on exit? |
| `flow_bias` | [−0.2, 0, +0.2] | One-sided flow impact? |
| `hedge_cds_on` | [True, False] | Value of hedging while waiting? |

Run $M \geq 200$ MC paths per configuration. Report distribution of P&L, Sharpe, ACR for both LP strategy and baseline. Highlight the crossover point where the baseline dominates.

# Complete Parameter Reference

| Parameter | Type | Default | Description |
|---|---|---|---|
| **Simulation** | | | |
| `T_days` | int | 60 | Horizon (trading days) |
| `dt_minutes` | float | 5 | Price step |
| `trading_hours` | float | 8 | Hours per day |
| `n_mc_paths` | int | 500 | Monte Carlo paths |
| `seed` | int | 42 | Random seed |
| **Price process** | | | |
| `p0` | float | 100.0 | Initial price |
| `sigma_daily_bps` | float | 50 | Daily vol |
| `kappa_daily` | float | 0.02 | Mean-reversion |
| `p_bar` | float | 100.0 | Long-run mean |
| `v_open` | float | 0.5 | Open vol bump |
| `v_close` | float | 0.3 | Close vol bump |
| `tau_v_hours` | float | 0.75 | Vol seasonality width |
| **Alpha signal** | | | |
| `IC` | float | 0.10 | Information coefficient |
| `IC_stress_mult` | float | 0.4 | IC degradation in stress |
| `alpha_horizon_days` | float | 10 | Signal horizon |
| `signal_refresh_hours` | float | 24 | Update frequency |
| `p_calm_to_stress` | float | 0.05 | Regime transition |
| `p_stress_to_calm` | float | 0.15 | Regime transition |
| **Position target** | | | |
| `gamma` | float | 1.0 | Risk aversion |
| `q_max` | int | 20 | Max position (lots) |
| `lot_size_mm` | float | 1.0 | $MM per lot |
| **RFQ arrivals** | | | |
| `rfq_rate_per_day` | float | 15 | Daily RFQ count |
| `A_open` | float | 0.8 | Open activity bump |
| `A_close` | float | 0.5 | Close activity bump |
| `tau_f_hours` | float | 0.75 | Seasonality width |
| `flow_bias` | float | 0.0 | Directional bias |
| `size_mu` | float | 0.5 | Log-mean size |
| `size_sigma` | float | 0.8 | Log-std size |
| `size_max` | int | 10 | Max trade size |

| Parameter | Type | Default | Description |
|---|---|---|---|
| `n_dealers_mean` | float | 4 | Mean competitors |
| `n_dealers_max` | int | 8 | Max competitors |
| `tox_a` | float | 2.0 | Toxicity Beta $a$ |
| `tox_b` | float | 8.0 | Toxicity Beta $b$ |
| **Observable mid / skew** | | | |
| `obs_lag_minutes` | float | 15 | Staleness |
| `obs_noise_bps` | float | 5 | Observation noise |
| `skew_accuracy` | float | 0.3 | Skew $\rho_s$ |
| **Lean** | | | |
| `lambda_base_bps` | float | 2.0 | Base lean |
| `sigma_ref_bps` | float | 50 | Reference vol |
| `kappa_urgency` | float | 1.0 | Urgency scale |
| `kappa_convexity` | float | 0.1 | Convexity per lot |
| `kappa_limit` | float | 10.0 | Soft limit penalty |
| `theta_limit` | float | 0.7 | Limit activation |
| **Competitor model** | | | |
| `dealer_bias_std_bps` | float | 3.0 | Cross-dealer dispersion |
| `markup_base_bps` | float | 8.0 | Baseline markup |
| `markup_N_bps` | float | $-1.0$ | Per-dealer shift |
| `markup_size_bps` | float | 2.0 | Per-log-lot shift |
| `markup_tox_bps` | float | 10.0 | Per-toxicity shift |
| `markup_noise_bps` | float | 3.0 | Per-quote noise |
| `quote_noise_bps` | float | 2.0 | Dealer model error |
| `respond_base` | float | 0.85 | Base response rate |
| `respond_size` | float | 0.05 | Size impact |
| `respond_tox` | float | 0.1 | Toxicity impact |
| `winrate_est_error` | float | 0.0 | Trader estimation error |
| `m_max_bps` | float | 30 | Markup grid extent |
| `m_grid_bps` | float | 0.1 | Grid resolution |
| **Post-fill / adverse selection** | | | |
| `adverse_move_bps` | float | 5.0 | Adverse move scale |
| **Hybrid exit** | | | |
| `aggress_window_hours` | float | 8 | Time before expiry to aggress |
| `aggress_halfspread_bps` | float | 8.0 | Half-spread cost |
| `aggress_impact_bps` | float | 2.0 | Impact per $\sqrt{\text{lot}}$ |
| **Street lean** | | | |
| `street_lean_mr` | float | 0.1 | Mean-reversion $\theta_b$ |

| Parameter | Type | Default | Description |
|---|---|---|---|
| `street_lean_vol_bps` | float | 2.0 | Vol $\sigma_b$ |
| `street_lean_eq` | float | 0.0 | Equilibrium |
| `street_obs_noise` | float | 0.5 | Proxy noise |
| `street_proxy_wts` | 3-vec | [.4,.4,.2] | Proxy weights |
| **Hedging** | | | |
| `hedge_cds_on` | bool | False | CDS toggle |
| `hedge_etf_on` | bool | False | ETF toggle |
| `hedge_tsy_on` | bool | False | Treasury toggle |
| `rho_cds` | float | 0.85 | CDS correlation |
| `rho_etf` | float | 0.70 | ETF correlation |
| `rho_tsy` | float | 0.50 | Treasury correlation |
| `cds_cost_bps` | float | 2.0 | CDS execution cost |
| `etf_cost_bps` | float | 1.0 | ETF execution cost |
| `tsy_cost_bps` | float | 0.5 | Treasury execution cost |
| **Carry** | | | |
| `coupon_bps` | float | 500 | Annual coupon |

End of Specification